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MATH 245A

Homework 2
Due: November 2, 2026

Problem 2.1

Let $B \subseteq \mathbb{R}^d$.
(a) Show that if μ is a Borel measure on $\mathbb{R}^d$ then $\mu_{|B}$ is also a Borel measure on $\mathbb{R}^d$.
(b) Show that if μ is a Borel regular measure on $\mathbb{R}^d$, if B is μ-measurable and of finite measure then $\nu := \mu_{|B}$ is a Radon measure on $\mathbb{R}^d$.

Solution.

Problem 2.2*

Suppose μ is an outer measure on X, $E_1, \cdots, E_n$ are disjoint subsets of X and $c_1, \cdots, c_n$ are distinct nonzero real numbers. Prove that $f = c_1\chi_{E_1} + \cdots + c_n\chi_{E_n}$ is a μ-measurable function if and only if $E_1, \cdots, E_n$ are μ-measurable sets.

Solution. $\Rightarrow$: Suppose f is measurable. Thus for any open set U, we know that $f^{-1}(U)$ is μ-measurable. Let

$$\delta = \min\big(\{|c_i| : 1 \le i \le n\} \cup \{|c_i - c_j| : 1 \le i, j \le n, i \ne j\}\big).$$

This is a minimum of a finite nonempty set of positive numbers (the c_i are nonzero and distinct), so $\delta > 0$. Thus $c_j \notin B(c_i, \delta)$ for $j \ne i$ and $0 \notin B(c_i, \delta)$. Note that $f^{-1}(B(c_i, \delta)) = f^{-1}(\{c_i\}) = E_i$, and thus E_i is μ-measurable. Thus each E_i is μ −measurable.

$\Longleftarrow$: Suppose that $E_1, ..., E_n$ are μ −measurable. We want to show f is μ −measurable, thus we want to show for any open set $U \subseteq \mathbb{R}$, $f^{-1}(U)$ is measurable. Let $U \subseteq R$ be a subset, let $A(U) := U \cap \{0\} \cup \{c_k : 1 \le k \le n, c_k \in U\}$. Note that for any subset $K \subseteq \mathbb{R}$, if $A(K) = \emptyset$, then $f^{-1}(K) = \emptyset$. Thus for a open set U where $A(U) = \emptyset \Rightarrow f^{-1}(U) = \emptyset$ which is μ −measurable. Note the following

$$\begin{aligned} f^{-1}(U) = f^{-1}(A(U) \cup (U - A(U))) &= f^{-1}(A(U)) \cup f^{-1}(U - A(U)) \\ &= f^{-1}(A(U)) \\ &= \begin{cases} \bigcup_{k \in A(U)} E_k \text{ if } 0 \notin A(U) \\ \left(\bigcup_{k=1}^{n} E_k\right)^c \cup \left(\bigcup_{k \in A(U)-\{0\}} E_k\right) \text{ if } 0 \in A(U) \end{cases} \end{aligned}$$

If $0 \in A(U)$, $\bigcup_{k=1}^{n} E_k$ μ −measurable (finite union) and its complement is μ measurable. Furthermore, $\cup_{k \in A(U)-\{0\}} E_k$ is a finite union and is μ −measurable. Finally the union of both sets is μ −measurable. If $0 \notin A(U)$, then $f^{-1}(U)$ is a finite union of measurable sets and is thus measurable.

Problem 2.3*

Suppose $X \subseteq \mathbb{R}$ is a Borel set and $f : X \to \mathbb{R}$ is a function such that $\{x \in X : f \text{ is not continuous at } x\}$ is a countable set. Prove that f is a Borel measurable function.

Solution. We will first show that any countable subset of $X \subseteq \mathbb{R}$ is Borel in X. Let $S = \{s_n : n \in \mathbb{N}\} \subseteq X$ be a countable subset of X.

$$S = \bigcup_{n \in \mathbb{N}} \{s_n\}$$

and since singletons are closed in $\mathbb{R}$, hence measurable, and S is a countable union of singletons, S is a Borel set in $\mathbb{R}$. Since $S = S \cap X \subseteq X$, it is also a Borel set of X. Let $D = \{x \in X : f \text{ is not continuous at } x\}$. By the prior argument, for any subset $S \subseteq D$, S is Borel measurable in X (implying that D itself is a Borel set). Note the following

$$f^{-1}(U) = (f|_{D^c})^{-1}(U) \cup (f\ |_D)^{-1}(U) \quad U \subseteq \mathbb{R}.$$

Note by assumption,

$$f|_{D^c} : D^c \to \mathbb{R}$$

is continuous. It suffices to check that for any open set U, $f^{-1}(U)$ is a Borel set. Note that since $f|_{D^c}$ is continuous, $(f|_{D^c})^{-1}(U)$ is a relatively open subset of D^c, thus there exists $V \subseteq X$ open (in subspace topology) such that

$$(f|_{D^c})^{-1}(U) = V \cap D^c.$$

V is a Borel set of X. Since D is a Borel set, D^c is also a Borel set (closed under complements). Thus $(f|_{D^c})^{-1}(U)$ is a Borel set. Furthermore,

$$(f|_D)^{-1}(U) \subseteq D \Rightarrow (f|_D)^{-1}(U) \text{ is a Borel Set.}$$

Thus $f^{-1}(U)$ is a Borel Set since the union of two Borel sets is a Borel set.

Problem 2.4*

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable at every element of $\mathbb{R}$. Prove that f' is a Borel measurable function from $\mathbb{R}$ to $\mathbb{R}$.

Solution. For all $y \in \mathbb{R}$, let $f_y(x) = f(x + y)$.

To show f' is Borel measurable, it suffices to show that $(f')^{-1}([r, \infty))$ is Borel measurable.

$$\begin{aligned}
(f')^{-1}([r,\infty)) &= \{x \in \mathbb{R} : f'(x) \geq r\} \\
&= \left\{x \in \mathbb{R} : \forall \varepsilon > 0, \exists \delta > 0 \ \text{ s.t. } \ \forall y \in (x-\delta, x+\delta) - \{x\}, \frac{f(y)-f(x)}{y-x} \geq r - \varepsilon\right\} \\
&= \bigcap_{q \in \mathbb{Q} \cap (0,\infty)} \bigcup_{\delta \in \mathbb{Q} \cap (0,\infty)} \{x \in \mathbb{R} : \forall y \in (-\delta, \delta) - \{0\}, (f(x+y) - f(x)) \geq (r-\varepsilon)y\} \\
&= \bigcap_{q \in \mathbb{Q} \cap (0,\infty)} \bigcup_{\delta \in \mathbb{Q} \cap (0,\infty)} \bigcap_{y \in \mathbb{Q} \cap (-\delta,\delta) - \{0\}} \{x \in \mathbb{R} :, (f(x+y) - f(x)) \geq (r-\varepsilon)y\}
\end{aligned}$$

The last equality follows from the continuity of f. Let $s \in (-\delta, \delta) - \{0\}$. Let q_n be the rational approximation of s. If we know that $(f(x+q_n) - f(x)) \geq (r-\varepsilon)q_n$ then taking limits doesn't affect the inequality and $(f(x+s) - f(x)) \geq (r-\varepsilon)s$. Thus we have,

$$(f')^{-1}([r,\infty]) = \bigcap_{q \in \mathbb{Q} \cap (0,\infty)} \bigcup_{\delta \in \mathbb{Q} \cap (0,\infty)} \bigcap_{y \in \mathbb{Q} \cap (-\delta,\delta) - \{0\}} \left(f_y - f\right)^{-1}([(r-\varepsilon)y, \infty))$$

Note that $f_y - f$ is continuous, so

$$\left(f_y - f\right)^{-1}([(r-\varepsilon)y, \infty)) \text{ is Borel.}$$

Hence $(f')^{-1}([r,\infty))$ is a borel set.

Problem 2.5*

Let h be a continuous and strictly increasing real valued function on $\mathbb{R}$. Prove that if $B \subseteq \mathbb{R}$, then $h(B)$ is a Borel set if and only if B is a Borel set.

Solution. Note the strict increasing property, enforces that f is injective. We will show that f is a open mapping. Let U be a open set in $\mathbb{R}$. Let $y \in f(U)$. By injectivity, there exists a unique $x \in U$ such that $f(x) = y$. By openness of U, there exists a $\delta > 0$ such that $(x-\delta, x+\delta) \subseteq U$. By strict monotonicity, $f(x-\delta) < y < f(x+\delta)$, thus $y \in (f(x-\delta, f(x-\delta)) = f((x-\delta, x+\delta)) \subseteq f(U)$. Thus $f(U)$ is open. Thus f is a open map.

Suppose $B \subseteq \mathbb{R}$, suppose $h(B)$ is a Borel set. It its a countable intersection or union of open sets. Since inverse image, respects unions and intersections, by continuity of h, B is also borel. Suppose B is Borel, it is also a countable union or intersection of open sets. Since taking the image also respects unions and intersections, and since h is a open map, $h(B)$ is also the countable intersection or union of open sets, and thus $h(B)$ is Borel.

Problem 2.6*

Suppose $f : B \to \mathbb{R}$ is a Borel measurable function. Define $g : \mathbb{R} \to \mathbb{R}$ to be the function which coincides with f on B and is identically equal to 0 on $\mathbb{R} \setminus B$. Prove that g is a Borel measurable function.

Solution. To prove that g is Borel, it suffices to show that $g^{-1}((r,\infty))$ is Borel $r \in \mathbb{Q}$ since $\{(r,\infty) : r \in \mathbb{Q}\}$ generate $\mathcal{B}(\mathbb{R})$. We can split it into two cases $r \geq 0$ and $r < 0$.

1. $r \geq 0$: Since $g(B^c) = \{0\}$, $B^c \cap g^{-1}((r,\infty)) \Rightarrow g^{-1}((r,\infty)) = f^{-1}((r,\infty))$, which is Borel measurable set inside B. Thus there exists a $U \in B(\mathbb{R})$ such that $f^{-1}((r,\infty)) = U \cap B$, and thus $g^{-1}((r,\infty)) = f^{-1}((r,\infty))$ is Borel measurable with respect to $\mathbb{R}$.
2. $r < 0$: $g^{-1}((r,\infty)) = B^c \cup f^{-1}((r,\infty))$. B^c is borel measurable because B is borel measurable and $\mathcal{B}(\mathbb{R})$ is closed under complements. $f^{-1}((r,\infty))$ is Borel measurable because of a argument identical to the argument in case 1.

Thus $g^{-1}((r,\infty))$ is always measurable for any choice of $r \in \mathbb{Q}$. Hence g is measurable.

Problem 2.7*

Say whether or not every open subset of $\mathbb{R}^d$ is a countable union of closed subsets of $\mathbb{R}^d$. Justify your answer (your argument should be short, say, half a page).

Solution. Let U be a open set. Thus U^c is a closed set. Note that the function

$$d(x) \coloneqq \inf_{a \in U^c} |x - a|$$

is continuous. Note that $\forall x \in U$, $d(x) > 0$. Assume the contrary that $d(x) = 0$. Thus $\exists a_n \in U^c$ such that $|x - a_n| \leq \frac{1}{n}$. Thus $a_n \to x$ and x is a limit point of elements of a closed set and thus $x \in U^c$, which is a contradiction. Thus $d(x) > 0$. Thus

$$U \subseteq \bigcup_{n \in \mathbb{N}} d^{-1}\left(\left[\frac{1}{n}, \infty\right)\right) = V.$$

Note that $\left[\frac{1}{n}, \infty\right) = \left(0, \frac{1}{n}\right)^c$ is closed thus $d^{-1}\left(\left[\frac{1}{n}, \infty\right)\right)$ is closed. Suppose $y \in V$, thus $\exists N \in \mathbb{N}$ such that $d(y) \geq \frac{1}{N}$, thus $y \in (U^c)^c = U$. Thus

$$V = U$$

Problem 2.8

Let μ be an outer measure on a nonempty set X. Let $f, g : X \to [-\infty, +\infty]$ be two μ-summable functions. Assume that $A, B \subseteq X$ are two μ-measurable sets. Recall

$$\int_A f \, \mathrm{d}\mu = \int_X \chi_A f \, \mathrm{d}\mu.$$

(a) Prove that f is $\mu_{|A}$-summable and $\int_X f \, \mathrm{d}\left(\mu_{|A}\right) = \int_X f \cdot \chi_A \, \mathrm{d}\mu$.
(b) Prove that $\int_X (f + g) \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu + \int_X g \, \mathrm{d}\mu$, and $\int_X cf \, \mathrm{d}\mu = c \int_X f \, \mathrm{d}\mu$, for all $c \in \mathbb{R}$.
(c) Prove that if $f \leq g$ then $\int_X f \, \mathrm{d}\mu \leq \int_X g \, \mathrm{d}\mu$.

Solution.

1. Let $g : X \to [0, \infty]$ be a μ-measurable simple function, i.e. with countable image $Y \subseteq [0, \infty]$. By definition,

$$\begin{aligned} \int_X g \, \mathrm{d}\left(\mu_{|A}\right) &= \sum_{y \in Y} y \, \mu_{|A}(g^{-1}\{y\}) \\ &= \sum_{y \in Y} y \, \mu(A \cap g^{-1}\{y\}). \end{aligned}$$

The function $g\chi_A$ is also simple, with image contained in $Y \cup \{0\}$, and $(g\chi_A)^{-1}\{y\} = A \cap g^{-1}\{y\}$ for every $y \neq 0$. The $y = 0$ term contributes 0 (with the convention $0 \cdot \infty = 0$), so

$$\int_X g\chi_A \, \mathrm{d}\mu = \sum_{y \in Y \setminus \{0\}} y \, \mu(A \cap g^{-1}\{y\}) = \sum_{y \in Y} y \, \mu(A \cap g^{-1}\{y\}).$$

Hence $\int_X g \,\mathrm{d}\big(\mu_{|A}\big) = \int_X g\chi_A \,\mathrm{d}\mu$. Suppose $g : X \to [-\infty, \infty]$ was a μ-measurable simple function. Then let $g^+ = \max(0, g)$ and $g^- = \max(0, -g)$ by the above we know that

$$\begin{aligned}\int_X g d(\mu|_A) &= \int_X g^+ \,\mathrm{d}(\mu|_A) - \int_X g^- \,\mathrm{d}(\mu|_A) \\ &= \int_X 1_A g^+ \,\mathrm{d}\mu - \int_X 1_A g^- \,\mathrm{d}\mu \\ &= \int_X 1_A g \,\mathrm{d}\mu\end{aligned}$$

Let $f : X \to [-\infty, \infty]$ be a μ-measurable function. There exists $(f_k)_{k=1}^\infty$ a sequence of simple functions f_k monotonically increasing to f.

$$\begin{aligned}\int_X f \,\mathrm{d}\big(\mu_{|A}\big) &= \int_X \lim_{k\to\infty} f_k \,\mathrm{d}\big(\mu_{|A}\big) \\ &= \lim_{k\to\infty} \int_X f_k \,\mathrm{d}\big(\mu_{|A}\big) \quad \text{(Monotone Convergence Theorem)} \\ &= \lim_{k\to\infty} \int_X \chi_A f_k \,\mathrm{d}\mu \\ &= \int_X \chi_A f \,\mathrm{d}\mu \quad \text{(Monotone Convergence Theorem).}\end{aligned}$$

Let $f : X \to [-\infty, \infty]$ a μ-measurable function. Let $f^+ = \max(0, f)$ and $f^- = \max(0, -f)$. Note that $f^+, f^- \geq 0$ and $f = f^+ - f^-$.

$$\int_X f \,\mathrm{d}\big(\mu_{|A}\big) = \int_X (f^+ - f^-) \,\mathrm{d}\big(\mu_{|A}\big)$$

Problem 2.9

Let μ be an outer measure on a nonempty set X, let $\{A_k\}_{k=1}^\infty$ be a collection of pairwise disjoint μ-measurable subsets of X, and let $f : X \to \mathbb{R}$ be a summable function. Prove that

$$\int_A f \,\mathrm{d}\mu = \sum_{k=1}^\infty \int_{A_k} f \,\mathrm{d}\mu,$$

where $A = \cup_{k=1}^\infty A_k$.

Solution.

Problem 2.10*

Let (Ω, μ) be an outer measure space and suppose $f_n, g_n, f, g \in L^1(\mu)$. Suppose

$$|f_n| \leq g_n, \quad f_n \to f \text{ a.e.}, \quad g_n \to g \text{ a.e.}, \quad \lim_{n\to\infty} \int_\Omega g_n(x)\mu(\mathrm{d}x) = \int_\Omega g(x)\mu(\mathrm{d}x).$$

Show that $\lim_{n\to\infty} \int_\Omega f_n(x)\mu(\mathrm{d}x) = \int_\Omega f(x)\mu(\mathrm{d}x)$.

Solution. Note that $|f| \le g$ a.e., since $|f_n| \to |f|$ and $g_n \to g$ a.e. and $|f_n| \le g_n$. Note the following

$$g_k + g - |f_k - f| \ge g_k + g - |f_k| - |f| \ge 0$$

Note the following fact about lim infs. Suppose $a_n \to a$ and b_n are sequences of real numbers.

$$\begin{aligned}\liminf_{n\to\infty}(a_n + b_n) &= \lim_{n\to\infty} \inf_{k\ge n}(a_k + b_k) \\ &\le \lim_{n\to\infty}\left(\sup_{k\ge n} a_k + \inf_{k\ge n} b_k\right) \\ &= a + \liminf_{n\to\infty} b_n\end{aligned}$$

and

$$\begin{aligned}\liminf_{n\to\infty}(a_n + b_n) &\ge \lim_{n\to\infty}\left(\inf_{k\ge n} a_k + \inf_{k\ge n} b_k\right) \\ &= a + \liminf_{n\to\infty} b_n.\end{aligned}$$

Thus

$$\liminf_{n\to\infty}(a_n + b_n) = a + \liminf_{n\to\infty} b_n.$$

Remember this fact, it will come later on in the proof. Thus we have…

$$\begin{aligned}\int_\Omega (2g)\,\mathrm{d}\mu &= \int_\Omega \lim_{k\to\infty}(g(x) + g_k(x))\,\mathrm{d}\mu \quad \text{(a.e. convergence of } g_k \to g\text{)} \\ &= \int_\Omega \liminf_{k\to\infty}(g_k(x) + g(x) - |f_k(x) - f(x)|)\,\mathrm{d}\mu \quad \text{(a.e. convergence of } f_k \to f\text{)} \\ &\le \liminf_{k\to\infty} \int_\Omega (g_k + g - |f_k - f|)\,\mathrm{d}\mu \quad \text{(Fatou's Lemma)} \\ &= \int_\Omega (2g)\,\mathrm{d}\mu + \liminf_{k\to\infty}\left(-\int_\Omega |f_k - f|\,\mathrm{d}\mu\right) \quad \text{(using lim inf fact above)} \\ &= \int_\Omega 2g\,\mathrm{d}\mu - \limsup_{k\to\infty} \int_\Omega |f_k - f|\,\mathrm{d}\mu\end{aligned}$$

Thus

$$\limsup_{n\to\infty} \int_\Omega |f_n - f|\,\mathrm{d}\mu \le 0 \Rightarrow \lim_{n\to\infty} \int_\Omega |f_n - f|\,\mathrm{d}\mu = 0 \Rightarrow \lim_{n\to\infty} \int_\Omega f_n\,\mathrm{d}\mu = \int_\Omega f\,\mathrm{d}\mu$$